Smart Home Surveillance System v2 - System Architecture

ESP32 firmware v2.2.0: safety logic on core 1, networking on core 0

SENSING (inputs)

PIR motion sensor
HC-SR501, output on GPIO13

Door / window switch
slide or reed switch, GPIO14
internal pull-up, HIGH = open

ARM / DISARM button
momentary, GPIO27
internal pull-up, LOW = pressed

ESP32 DevKit (dual core, FreeRTOS)

Core 1: safety loop (Arduino loop task)
never blocks, watchdog-supervised, works offline

1. Input conditioning
debounce 40 ms on door and button
PIR: 10 s warm-up, 150 ms confirm

2. State machine
DISARMED > EXIT_DELAY 10 s > ARMED
ARMED + motion / door > ENTRY_DELAY 10 s
ENTRY_DELAY expires > ALARM 30 s > ARMED
button arms / disarms; remote: ARM, DISARM, STATUS
armed state kept in NVS across power cuts

3. Output driver
buzzer tone 2.7 kHz, 2 LEDs, OLED at 4 Hz

Command
queue, 4 deep

Event queue
16 deep
drop oldest

Core 0: network task
slow DNS / TCP can never stall the alarm

Wi-Fi manager
retry every 15 s, NTP clock (IST)

MQTT client
Last Will "offline", reconnect every 5 s
subscribes to .../cmd
Home Assistant discovery

publish

Event publisher
JSON events with sequence no., oldest first
retained state + 30 s heartbeat

LOCAL OUTPUTS

Buzzer
passive piezo
GPIO23

Status LEDs
green GPIO26
red GPIO25

OLED SSD1306
I2C 0x3C
GPIO21 / 22

NETWORK AND USER SIDE

Wi-Fi, MQTT 1883 / 8883 TLS

MQTT broker
demo: broker.hivemq.com (public)
topics: availability, state, event, cmd
deployment: private broker, TLS + login

event, state

ARM, STATUS

Homeowner clients
Home Assistant / Node-RED / dashboard
phone MQTT app: alert on "alarm" event

LEGEND

- Sensing
- Safety core (core 1)
- Network core (core 0)
- Shared queues
- Local outputs
- Cloud / user

Events raised while offline wait in the queue (up to 16) and are sent on reconnect.

Remote DISARM needs a valid HMAC tag over a fresh, NVS-persisted counter.